



## **Energy Use and Sulphur Dioxide Emissions in Asia**

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This paper presents a review of energy use in 22 selected countries of Asia and estimates the anthropogenic emission of sulphur dioxide (SO<sub>2</sub>) for the selected countries, both at national and disaggregated sub-country regional levels. The paper also makes a comparative assessment of the Asian countries in terms of SO<sub>2</sub> emission intensity (i.e. emission per GDP), emission per capita and emission density (i.e. emission per unit area).

Total SO<sub>2</sub> emission in the region was estimated to be about 38 million tons in 1990. Five countries, China, India, South Korea, Japan and Thailand, accounted for over 91% of the regional SO<sub>2</sub> emission. Coal use had the dominant share (81%) of the total emission from the region. Among the economic sectors, industry contributed the largest share (49%) to the total emissions of the selected countries as a whole, followed by the power sector (30%). These findings suggest the need for mitigation strategies focussed on the industry and power sectors of the major emitting countries in Asia.

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### **1. Introduction**

Although, until recently, acidification of the environment has been regarded as a problem only in Europe and North America, the problem has also started to emerge in parts of Asia. Most of the air pollution problems identified in the Asian countries are confined to localised areas of high emissions (usually associated with the densely populated cities of the region) such as Beijing, Bombay, Calcutta, Shenyang, Seoul, Bangkok, Hong Kong, Manila and Kuala Lumpur (Rodhe *et al.*, 1988; Bhatti *et al.*, 1989; World Bank Report, 1991; IIASA, 1993). Pollutant concentrations in most of these cities regularly exceed the World Health Organization (WHO) recommended limits for sulphur dioxide (SO<sub>2</sub>). The Asian region comprises many of the fastest growing economies in the world. During the period of 1971–1991, the average total primary energy consumption in Asia has been growing at the rate of 5.9% per annum,

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as compared to the world average of 2.4% (OECD/IEA, 1993). The demand for electricity is growing even faster. In particular, electricity generation from thermal plants in 16 developing member countries of the Asian Development Bank (ADB) taken as a whole, has grown at an annual rate of 8.9% during 1973–1990 (ADB, 1992). Thus, from the regional environmental perspective, the growth of energy in Asia is likely to assume an increasingly important role in the future.

As is well known, the anthropogenic emission of  $\text{SO}_2$  is the major precursor of acid rain. Reliable estimation of anthropogenic emissions of acid rain precursors is important in order to understand their geographical origins. Furthermore, assessment of region-specific emissions by economic sector and fuel type is necessary in order to be able to formulate effective strategies for the mitigation of these precursors.

There exist a limited number of studies that have estimated the inventories of  $\text{SO}_2$  emissions in Asian countries (Bhatti *et al.*, 1989; Foell and Green, 1990; Kato and Akimoto, 1992; NISTEP, 1993). Except Kato and Akimoto (1992) and NISTEP (1993), the studies are based on national level data and thus fail to provide the spatial distribution of emissions across different regions of individual countries. The studies by Kato and Akimoto (1992) and NISTEP (1993) have provided emission estimates only at the national level, except for China and India. The distinctive feature of the present paper is that emission levels are estimated at sub-country regional levels for all major countries of Asia using region-specific data that clearly account for variation in fuel qualities. The study also analyses sectoral contributions to total emissions and assesses emissions from the selected countries in terms of indicators such as per capita emissions, emission intensity (defined as emission per unit of GDP) and emission density (defined as emission per unit of area). Furthermore, the emission inventory in our study is for a more recent year (1990) as compared to the previous studies.

This study covers 22 countries in Asia, China, Japan, Hong Kong, South Korea, Mongolia and Taiwan in north-east Asia, Bangladesh, Bhutan, India, Myanmar, Pakistan, Nepal and Sri Lanka in south Asia, and Brunei, Cambodia, Indonesia, Laos, Malaysia, Philippines, Singapore, Thailand and Vietnam in south-east Asia. The national level emission estimates were derived from estimates of the emissions at the sub-country regional levels. Altogether, the 22 countries were disaggregated into 92 sub-country regions for this purpose.

## 2. Methodology

Total anthropogenic emission of  $\text{SO}_2$  in a country or region comprises emissions from two sources, i.e. combustion of fuels and non-energy use of fuels in industrial processes. The emission of  $\text{SO}_2$  from fuel combustion is estimated by taking into account sulphur content and heat value of fuels as well as the efficiency of emission control (wherever applicable) and sulphur retained in ash.

To present the underlying methodology for calculation of the  $\text{SO}_2$  emissions, the following symbols are defined:

|              |                                                                                           |
|--------------|-------------------------------------------------------------------------------------------|
| $S_{ijk}(t)$ | = $\text{SO}_2$ emission by fuel type $i$ used in sector $j$ of country $k$ in year $t$ ; |
| $E_{ijk}(t)$ | = Consumption of fuel type $i$ used in sector $j$ of country $k$ in year $t$ ;            |
| $SC_{ijk}$   | = Sulphur content of fuel type $i$ used in sector $j$ of country $k$ ;                    |
| $hv_{ij}$    | = Heat value of fuel type $i$ used in sector $j$ ;                                        |

|            |                                                                                                     |
|------------|-----------------------------------------------------------------------------------------------------|
| $sr_{jk}$  | = Fraction of SO <sub>2</sub> emissions retained in ash in sector $j$ of country $k$ ;              |
| $x_{jk}$   | = Fraction of emissions removed by pollution control in sector $j$ of country $k$ ; and             |
| $S_i^P(t)$ | = SO <sub>2</sub> emissions from industrial non-combustion processes of fuel type $i$ in year $t$ . |

SO<sub>2</sub> emission through the use of fuel  $i$  in sector  $j$  of country  $k$  can be expressed as:

$$S_{ijk}(t) = \sum_i E_{ijk}(t) (sc_{ijk}/hv_{ij}) (1 - sr_{jk}) (1 - x_{jk}) \quad (1)$$

The total emission in country  $k$  is given by:

$$S_k(t) = \sum_i \sum_j S_{ijk}(t) + \sum_i S_i^P(t) \quad (2)$$

SO<sub>2</sub> emissions were estimated both at the country and sub-country regional levels. In total, 17 kinds of fuels and 5 major economic sectors were considered for the emission analysis.

### 2.1. DATA SOURCES

The estimation of SO<sub>2</sub> emissions in the present study is based on a spatially disaggregated energy and socio-economic database which was established at the Asian Institute of Technology (AIT) in the framework of a regional project entitled "Acid Rain and Emissions Reduction in Asia", funded by the ADB. The database has been created on the basis of data supplied by national collaborating institutes/experts of countries, as well as on data from published sources such as ADB (1992), ADB (1993), OECD/IEA (1993), NISTEP (1993), AEEMTRC (1993) and relevant country-specific publications.

## 3. Energy use in Asia

In recent years, the Asian region has recorded the highest growth rate of total primary commercial energy (TPCE) Consumption (BP, 1994). The region accounted for about 23.5% of the TPCE consumption in the world in 1993 (*ibid.*). About 84% of the total TPCE consumption in Asia was accounted for by four countries, China, India, Japan and South Korea combined.

Currently, energy use in Asia as a whole is heavily dependent on fossil fuels (almost 93% of the TPCE consumption was from fossil fuel in 1993). About 46% of the TPCE consumption in 1993 was in the form of coal (mainly due to high coal consumption in China and India) followed by oil (39%), natural gas (7.5%) and primary electricity (7.5%). Thus, coal and oil contribute significantly to the emission of acid rain precursors in Asia.

In per capita terms, commercial energy consumption in the Asian region as a whole was about 0.5 tons of oil equivalent (toe) (which was less than one tenth of that in OECD) in 1990. However, the average per capita energy consumption in the region has been growing rapidly as it almost doubled during 1973–1990 (ADB, 1992). Although the average energy consumption per capita figure is relatively low, the evolution of the energy sector in Asia has important environmental implications, mainly for two reasons. First, in many Asian countries, energy use is growing at a higher rate than in most countries in other parts of the world. Second, most Asian countries have experienced significant changes in their energy use pattern in the recent years. In particular, with

TABLE 1. Average annual growth rate of fossil fuels, primary electricity and total primary energy consumption (TPEC) during 1983–93 (%)

| Region            | Coal | Oil  | Gas | Nuclear | Hydro | TPEC |
|-------------------|------|------|-----|---------|-------|------|
| World             | 1.1  | 1.2  | 3.0 | 8.0     | 2.0   | 1.9  |
| Asia              | 4.4  | 4.7  | 9.3 | 9.9     | 4.4   | 5.0  |
| North America     | 2.0  | 1.1  | 2.0 | 7.5     | 0.0   | 1.9  |
| Latin America     | 3.4  | 1.8  | 2.2 | 13.5    | 4.1   | 2.9  |
| OECD Europe       | –1.6 | 1.0  | 3.6 | 8.1     | 0.9   | 1.6  |
| Non-OECD Europe   | –3.3 | –4.0 | 2.2 | 6.5     | 2.6   | –1.1 |
| Rest of the world | 1.7  | 2.6  | 6.3 | 6.9*    | 4.0   | 3.2  |

\* During the years 1984–1993.

Source: BP (1994).

rapid industrialisation in many countries, there has been an increasing switch from traditional to modern fuels. Furthermore, as can be seen from Table 1, fossil fuel use (i.e. of coal, oil, natural gas) has registered faster growth rates than in the rest of the world.

### 3.1. SECTORAL SHARES

Sectoral shares of total energy use are largely determined by the extent of industrialisation and urbanisation in a country. Table 2 presents the sectoral shares in total primary energy consumption in the Asian countries in 1990. Note here that the “domestic” sector in the table consists of residential, commercial and agricultural sectors. As can be seen from the table, the share of the industrial sector in total primary energy consumption was the highest in the selected countries as a whole, followed by the domestic, power generation and transport sectors. However, this was not always the case at the individual country or sub-country regional levels: the industrial sector was found to have the largest share in total energy consumption in north-east Asia while the domestic sector dominated emissions from the south and south-east Asian regions. The value of the industrial sector share also varied widely across the countries. It was the highest in China, with over 60% of total energy consumption in that country, while it was in the range of 30–44% in Japan, South Korea, Malaysia, Philippines, Singapore and Taiwan.

The share of the domestic sector in total energy consumption ranged from 9–95%. The share was found to be relatively high (in the range of 34–95% in Bangladesh, Bhutan, India, Indonesia, Laos, Myanmar, Nepal, Sri Lanka, Thailand and Vietnam. The power sector had the largest share in total primary energy consumption in Brunei, Hong Kong and Mongolia and the second largest in Japan, Singapore and Taiwan. Transport sector shares were relatively small except in Thailand, Hong Kong, Brunei, Philippines and Malaysia.

### 3.2. ENERGY MIX

Table 3 presents the fuel shares of TPCE in the Asian countries in 1990. The energy mix in the selected countries as a whole was dominated by coal, followed by oil, primary electricity and gas. Coal had the largest share in the fuel mix in both north-east and

TABLE 2. Sectoral shares in total primary energy consumption of Asian countries in 1990 (%)

| Region/country         | Industry    | Domestic*   | Transport   | Power       | Non-energy |
|------------------------|-------------|-------------|-------------|-------------|------------|
| <b>North-east Asia</b> | <b>50.5</b> | <b>19.2</b> | <b>8.8</b>  | <b>17.6</b> | <b>3.8</b> |
| China                  | 60.3        | 18.7        | 3.5         | 15.3        | 2.2        |
| Hong Kong              | 13.7        | 22.3        | 22.7        | 41.3        | 0.0        |
| Japan                  | 37.1        | 19.4        | 16.3        | 19.9        | 7.3        |
| Mongolia               | 9.2         | 35.2        | 5.0         | 49.7        | 0.8        |
| South Korea            | 37.7        | 25.9        | 16.5        | 19.9        | 0.0        |
| Taiwan                 | 41.7        | 13.7        | 13.7        | 23.5        | 7.4        |
| <b>South Asia</b>      | <b>30.4</b> | <b>39.2</b> | <b>9.4</b>  | <b>19.7</b> | <b>1.3</b> |
| Bangladesh             | 20.9        | 56.2        | 3.6         | 9.7         | 9.6        |
| Bhutan                 | 0.8         | 95.2        | 0.6         | 3.4         | 0.0        |
| India                  | 33.6        | 35.2        | 9.6         | 21.6        | 0.0        |
| Myanmar                | 5.8         | 79.9        | 4.7         | 8.3         | 1.3        |
| Nepal                  | 2.8         | 93.8        | 1.7         | 1.8         | 0.0        |
| Pakistan               | 26.6        | 35.5        | 12.4        | 18.5        | 6.0        |
| Sri Lanka              | 11.2        | 59.4        | 17.3        | 10.3        | 1.8        |
| <b>South-east Asia</b> | <b>23.7</b> | <b>44.3</b> | <b>17.5</b> | <b>13.2</b> | <b>1.3</b> |
| Brunei                 | 7.9         | 14.4        | 31.2        | 45.7        | 0.9        |
| Cambodia               | 19.8        | 52.5        | 15.8        | 11.9        | 0.0        |
| Indonesia              | 18.1        | 60.0        | 14.5        | 7.0         | 0.4        |
| Laos                   | 4.7         | 88.6        | 1.6         | 5.1         | 0.0        |
| Malaysia               | 32.2        | 18.4        | 22.5        | 19.3        | 7.6        |
| Philippines            | 29.8        | 16.6        | 24.6        | 27.6        | 1.3        |
| Singapore              | 43.9        | 9.8         | 17.7        | 28.6        | 0.0        |
| Thailand               | 25.8        | 34.2        | 23.8        | 16.0        | 0.1        |
| Vietnam                | 21.6        | 64.1        | 5.7         | 8.2         | 0.6        |
| <b>Asia (total)</b>    | <b>44.2</b> | <b>25.4</b> | <b>9.8</b>  | <b>17.5</b> | <b>3.1</b> |

\* Comprises residential, commercial and agricultural sectors.

south Asia. It is interesting to note that the share of oil in south-east Asia was much higher (more than 72%) than in the other parts of the region, partly due to oil consumption in the Association of South-east Asian Nation (ASEAN) countries (i.e. Brunei, Indonesia, Malaysia, Philippines, Singapore and Thailand). Four countries, China, India, South Korea and Japan accounted for over 80% of the total coal consumption in the Asian region as a whole in 1990. Furthermore, the shares of these countries in the total regional oil and gas consumption were more than 70% and 74%, respectively.

### 3.3. ENERGY INTENSITIES

Energy intensities (defined as energy used per unit of GDP) of selected Asian countries are shown in Table 4, together with those of some western, industrialised economies. The intensity was found to have an increasing trend in countries like Indonesia and Malaysia and a decreasing trend in China, Japan, South Korea, Philippines and Thailand during 1980–1990. As can be seen from the table, energy intensity varied widely across the countries. It ranged from 0.258 toe/thousand US\$ in Japan to 1.659 toe/thousand US\$ in China in 1990. The energy intensities of some major countries like China, India, South Korea, Indonesia, Malaysia, Philippines and Thailand were found to be higher than the average intensity of OECD countries.

TABLE 3. Fuel shares in TPCE of Asian countries in 1990 (%)

| Region/country         | Coal        | Oil         | Gas         | Primary electricity |
|------------------------|-------------|-------------|-------------|---------------------|
| <b>North-east Asia</b> | <b>55.5</b> | <b>29.7</b> | <b>6.3</b>  | <b>8.5</b>          |
| China                  | 84.0        | 10.4        | 1.7         | 3.9                 |
| Hong Kong              | 64.4        | 32.0        | 3.5         | —                   |
| Japan                  | 12.2        | 56.6        | 15.4        | 15.8                |
| South Korea            | 26.4        | 52.4        | 3.9         | 17.4                |
| Mongolia               | 58.3        | 22.5        | 19.2        | —                   |
| Taiwan                 | 28.2        | 65.1        | 2.4         | 4.3                 |
| <b>South Asia</b>      | <b>50.0</b> | <b>28.8</b> | <b>11.7</b> | <b>9.6</b>          |
| Bangladesh             | 5.8         | 30.2        | 60.0        | 3.9                 |
| Bhutan                 | 3.7         | 27.5        | 68.8        | —                   |
| India                  | 58.1        | 27.8        | 5.5         | 8.6                 |
| Myanmar                | 1.6         | 34.9        | 46.8        | 16.7                |
| Nepal                  | 11.2        | 54.9        | —           | 33.9                |
| Pakistan               | 8.0         | 31.8        | 44.8        | 15.4                |
| Sri Lanka              | 0.1         | 63.1        | —           | 36.8                |
| <b>South-east Asia</b> | <b>9.9</b>  | <b>72.5</b> | <b>13.2</b> | <b>4.4</b>          |
| Brunei                 | —           | 40.5        | 59.5        | —                   |
| Cambodia               | 44.2        | 46.2        | —           | 9.6                 |
| Indonesia              | 8.1         | 77.0        | 12.4        | 2.4                 |
| Laos                   | —           | 43.4        | —           | 56.6                |
| Malaysia               | 5.9         | 71.3        | 18.8        | 4.0                 |
| Philippines            | 8.0         | 72.3        | 10.4        | 9.3                 |
| Singapore              | 0.1         | 99.1        | 0.9         | —                   |
| Thailand               | 14.2        | 64.8        | 17.0        | 4.0                 |
| Vietnam                | 32.0        | 52.0        | 0.1         | 16.0                |
| <b>Asia (total)</b>    | <b>51.1</b> | <b>33.1</b> | <b>7.6</b>  | <b>8.3</b>          |

#### 4. Emission of SO<sub>2</sub> in Asia

##### 4.1. EMISSIONS BY COUNTRY AND SUB-COUNTRY REGIONS

Table 5 presents the estimated emissions of SO<sub>2</sub> in Asia both at national and sub-country regional levels in 1990. The total emissions of SO<sub>2</sub> in the region were estimated to be about 38 million tons in 1990, almost 56% higher than that of North America (United States and Canada). With the high growth of fossil fuel use in Asia, total SO<sub>2</sub> emission is expected to become significantly higher in the coming decades. As can be seen from Table 5, countries in north-east Asia as a group (China, Hong Kong, Japan, South Korea, Mongolia and Taiwan) contributed almost 78% of the total emissions of SO<sub>2</sub> in Asia, followed by the south Asia region (15%) and south-east Asia region (7%). The SO<sub>2</sub> emissions were found to be relatively high in most of the coal-dominated countries. China had the highest share (69%) of Asia's total SO<sub>2</sub> emission, while China, India, South Korea, Thailand and Japan together accounted for over 91% of the total.

Low SO<sub>2</sub> emission (below 30 thousand tons in 1990) from Bhutan, Cambodia, Laos, Myanmar and Nepal can be attributed mainly to the predominance of traditional (biomass) fuels, the sulphur contents of which are much lower than that of fossil fuels. High dependence on traditional fuels was also partly responsible for relatively low SO<sub>2</sub> emissions in Bangladesh, Indonesia, Mongolia, Myanmar, Philippines, Sri Lanka and Vietnam.

TABLE 4. Energy intensity of selected Asian countries and western industrialised economies (toe/thousand US\$ at 1985 price)

| Region/country | 1973  | 1980  | 1985  | 1990  |
|----------------|-------|-------|-------|-------|
| Bangladesh*    | 0.245 | 0.293 | 0.331 | 0.450 |
| China*         | —     | 2.388 | 1.867 | 1.659 |
| India*         | 0.570 | 0.638 | 0.656 | 0.673 |
| Japan§         | 0.370 | 0.310 | 0.269 | 0.258 |
| Indonesia*     | 0.283 | 0.394 | 0.440 | 0.459 |
| South Korea*   | 0.596 | 0.688 | 0.602 | 0.605 |
| Malaysia*      | 0.375 | 0.376 | 0.394 | 0.414 |
| Myanmar*       | 0.355 | 0.323 | 0.318 | 0.279 |
| Nepal*         | 0.296 | 0.116 | 0.139 | 0.148 |
| Pakistan*      | 0.529 | 0.588 | 0.633 | 0.744 |
| Philippines*   | 0.458 | 0.390 | 0.366 | 0.400 |
| Singapore*     | 0.497 | 0.433 | 0.781 | 0.427 |
| Sri Lanka*     | 0.337 | 0.282 | 0.305 | 0.299 |
| Thailand*      | 0.480 | 0.433 | 0.424 | 0.479 |
| United States§ | 0.563 | 0.511 | 0.441 | 0.419 |
| France§        | 0.417 | 0.393 | 0.384 | 0.364 |
| North America§ | 0.570 | 0.521 | 0.450 | 0.428 |
| OECD Europe§   | 0.525 | 0.472 | 0.446 | 0.403 |

\* The GDP figures are based on IMF (1993) and IMF (1994); energy consumption figures are based on ADB (1993) and OECD/IEA (1993).

§ The figures are based on OECD/IEA (1994).

Some regions of China (Hebei-Anhui-Henah, Jiangsu, Sichuan and Heilongjiang-Jilin-Liaoning) emitted more than two million tons of SO<sub>2</sub> in 1990—a figure much higher than that of other Asian countries except India and South Korea (refer to Table 8). Among the regions in China, emission was the highest in Jiangsu, accounting for 13% of the total emissions from the Asian region as a whole.

Table 6 presents emission intensity, emission per capita and emission density for countries/regions in 1990. Emission intensity was found to vary from 0.3 kg/thousand US\$ (1990) in Japan to 71 kg/thousand US\$ in China. Emission intensities were particularly high in China and Mongolia among the countries in north-east Asia, India and Pakistan among those in south Asia, and Thailand and Vietnam in south-east Asia. As can be seen from the table, emission intensities in most Asian countries are higher than those in industrialised countries. These higher values in most Asian countries partly reflect low efficiency in energy use in most of the major countries. For example, as can be seen from Table 7, specific energy consumption figures (defined as energy consumption per ton of output) of major energy intensive industries in China and India were significantly higher than those in Japan, Sweden and the USA.

Furthermore, efficiency of electricity generation from coal-fired power plants was significantly lower in China (31.3%) and India (28.4%), compared to 38.8% in Japan and 36.9% in OECD countries as a whole (ADB, 1993; OECD/IEA, 1994). This partly reflects relatively high levels of pollution from the energy intensive production sector and the lack of emission controls. It is likely that the emission intensities in China, India, Thailand, Pakistan and Vietnam have been increasing during 1980–1990, since overall energy intensities and the use of coal have been increasing in these countries during the period.

TABLE 5. Emission of SO<sub>2</sub> in Asian and selected western countries in 1990 (thousand tons)

| Country            | Region               | Emission        | Country            | Region                  | Emission        |
|--------------------|----------------------|-----------------|--------------------|-------------------------|-----------------|
| <b>Bangladesh</b>  |                      | <b>87·4</b>     | <b>Pakistan</b>    |                         | <b>620·6</b>    |
|                    | Dhaka                | 13·4            |                    | Karachi                 | 105·5           |
|                    | Rest of country      | 74·1            |                    | Lahore                  | 21·9            |
| <b>Bhutan</b>      | <b>Whole country</b> | <b>2·1</b>      |                    | North and West Plateau  | 103·5           |
| <b>Brunei</b>      | <b>Whole country</b> | <b>3·9</b>      |                    | Punjab                  | 289·0           |
| <b>Cambodia</b>    | <b>Whole country</b> | <b>19·6</b>     |                    | Sind                    | 100·8           |
| <b>China§</b>      |                      | <b>26 373·9</b> | <b>Philippines</b> |                         | <b>414·9</b>    |
| <b>India§</b>      |                      | <b>4951·5</b>   |                    | Central Island-Mindanao | 93·2            |
| <b>Hong Kong</b>   | <b>Whole country</b> | <b>138·5</b>    |                    | Luzon                   | 176·0           |
| <b>Indonesia</b>   |                      | <b>691·1</b>    |                    | Metro Manila            | 145·7           |
|                    | Jakarta              | 23·1            | <b>Singapore</b>   | <b>Whole country</b>    | <b>190·9</b>    |
|                    | Java                 | 411·4           | <b>Sri Lanka</b>   | <b>Whole country</b>    | <b>30·5</b>     |
|                    | Rest of the country  | 122·8           | <b>Taiwan</b>      | <b>Whole country</b>    | <b>512·6</b>    |
|                    | Sumatra              | 133·8           | <b>Thailand</b>    |                         | <b>1036·9</b>   |
| <b>Japan</b>       |                      | <b>852·1</b>    |                    | Bangkok                 | 298·1           |
|                    | Chugoku-Shikoku      | 161·1           |                    | Central Valley          | 129·6           |
|                    | Chubu                | 162·3           |                    | North-east Plateau      | 58·9            |
|                    | Hokkaido-Tohoku      | 109·8           |                    | North Highlands         | 490·9           |
|                    | Kanto                | 178·8           |                    | South Peninsula         | 59·3            |
|                    | Kinki-Kyoto          | 134·3           | <b>Vietnam</b>     |                         | <b>134·4</b>    |
|                    | Kyushu-Okinawa       | 105·9           |                    | North Vietnam           | 62·8            |
| <b>South Korea</b> |                      | <b>1770·2</b>   |                    | South Vietnam           | 71·6            |
|                    | North                | 257·0           | <b>Asia</b>        |                         | <b>38 146·2</b> |
|                    | Pusan                | 661·3           |                    | North-east Asia         | 29 694·0        |
|                    | Seoul-Inchon         | 526·9           |                    | South Asia              | 5742·2          |
| <b>Laos</b>        | <b>Whole country</b> | <b>4·2</b>      |                    | South-east Asia         | 2710·0          |
| <b>Malaysia</b>    |                      | <b>214·3</b>    |                    | Asia                    |                 |
|                    | Kuala Lumpur         | 11·7            | Canada*            |                         | 3326·0          |
|                    | Peninsular Malaysia  | 168·6           | Germany*           |                         | 5679·0          |
|                    | Sarawak-Sabah        | 34·0            | United Kingdom*    |                         | 3780·0          |
| <b>Mongolia</b>    | <b>Whole country</b> | <b>46·7</b>     | United States*     |                         | 21 060·0        |
| <b>Myanmar</b>     | <b>Whole country</b> | <b>22·1</b>     |                    |                         |                 |
| <b>Nepal</b>       | <b>Whole country</b> | <b>28·0</b>     |                    |                         |                 |

\* These figures are based on WRI (1994).

§ Refer to Table 8 for sub-country regional emissions of SO<sub>2</sub>.

In per capita terms, SO<sub>2</sub> emission varied from 0·5 kg in Myanmar to 70 kg in Singapore. In most of the countries, the figure was below 25 kg, which was less than the corresponding figures in industrialised countries like the U.S.A., United Kingdom and Germany (WRI, 1994). The relatively low emission per capita figures for most Asian countries reflect the relatively lower level of industrialisation in these countries.

The average SO<sub>2</sub> emission density of the Asian region as a whole was estimated to be 1·86 tons/km<sup>2</sup>. Although the Asian regional average appears low compared to industrial countries like France, Germany, U.S.A. and the United Kingdom, there were

TABLE 6. SO<sub>2</sub> emission intensity, emission per capita and emission density in Asian and selected western countries in 1990

| Region/country         | Emission intensity<br>(kg/thousand US\$,<br>1990 price) | Emission per<br>capita<br>(kg) | Emission density<br>(tons/km <sup>2</sup> ) |
|------------------------|---------------------------------------------------------|--------------------------------|---------------------------------------------|
| <b>North-east Asia</b> | <b>7·88</b>                                             | <b>22·03</b>                   | <b>2·58</b>                                 |
| China                  | 71·35                                                   | 22·86                          | 2·79                                        |
| Hong Kong              | 3·45                                                    | 23·88                          | 138·50                                      |
| Japan                  | 0·29                                                    | 6·90                           | 2·25                                        |
| South Korea            | 6·98                                                    | 41·29                          | 17·88                                       |
| Mongolia               | 24·99                                                   | 21·93                          | 0·03                                        |
| Taiwan                 | 3·35                                                    | 25·38                          | 14·24                                       |
| <b>South Asia</b>      | <b>14·39</b>                                            | <b>5·07</b>                    | <b>1·11</b>                                 |
| Bangladesh             | 4·10                                                    | 0·77                           | 0·61                                        |
| Bhutan                 | 7·23                                                    | 1·33                           | 0·04                                        |
| India                  | 16·29                                                   | 5·99                           | 1·50                                        |
| Myanmar                | 0·96                                                    | 0·53                           | 0·03                                        |
| Nepal                  | 9·05                                                    | 1·48                           | 0·20                                        |
| Pakistan               | 15·74                                                   | 5·54                           | 0·78                                        |
| Sri Lanka              | 3·79                                                    | 1·79                           | 0·46                                        |
| <b>South-east Asia</b> | <b>8·18</b>                                             | <b>6·63</b>                    | <b>0·71</b>                                 |
| Brunei                 | 1·87                                                    | 15·26                          | 0·64                                        |
| Cambodia               | 12·22                                                   | 2·35                           | 0·11                                        |
| Indonesia              | 6·51                                                    | 3·85                           | 0·36                                        |
| Laos                   | 4·84                                                    | 1·01                           | 0·02                                        |
| Malaysia               | 5·02                                                    | 12·07                          | 0·65                                        |
| Philippines            | 9·40                                                    | 6·78                           | 1·38                                        |
| Singapore              | 5·23                                                    | 70·44                          | 190·89                                      |
| Thailand               | 12·93                                                   | 18·49                          | 2·02                                        |
| Vietnam                | 10·37                                                   | 2·03                           | 0·41                                        |
| <b>Asia</b>            | <b>8·47</b>                                             | <b>13·19</b>                   | <b>1·86</b>                                 |
| United States          | 3·81                                                    | 84·27                          | 2·25                                        |
| Germany                | 3·76                                                    | 89·96                          | 15·96                                       |
| France                 | 1·00                                                    | 21·15                          | 2·17                                        |
| United Kingdom         | 3·84                                                    | 65·84                          | 15·43                                       |

Sources: World Tables (1992), ADB (1993), IMF (1994) and AIT (1995).

TABLE 7. Specific energy consumption of industries in selected countries (million kCal/ton)

| Country        | Steel | Cement | Pulp and paper | Fertilizer |
|----------------|-------|--------|----------------|------------|
| China          | 6·87  | 2·24   | —              | 15·60      |
| India          | 9·50  | 2·00   | 11·13          | 11·25      |
| Japan          | 4·18  | 1·20   | —              | —          |
| Germany        | 5·21  | 0·82   | —              | —          |
| Italy          | 4·03  | 0·89   | —              | 9·92       |
| Sweden         | 5·02  | 1·40   | 7·56           | —          |
| United Kingdom | 6·07  | 1·30   | 7·62           | 12·23      |
| U.S.A.         | 6·06  | 0·95   | 9·70           | 11·32      |

Source: Chen *et al.* (1994).

TABLE 8. Emission density of SO<sub>2</sub> in selected sub-country regions of China and India in 1990

| Region/country               | SO <sub>2</sub> emission<br>(thousand tons) | Area*<br>(thousand km <sup>2</sup> ) | Emission density<br>(tons/km <sup>2</sup> ) |
|------------------------------|---------------------------------------------|--------------------------------------|---------------------------------------------|
| <b>China</b>                 | <b>26 373.9</b>                             | <b>9444.3</b>                        | <b>2.8</b>                                  |
| Beijing                      | 313.2                                       | 16.8                                 | 18.6                                        |
| Chongqing                    | 1023.5                                      | —                                    | —                                           |
| Fujian                       | 316.4                                       | 120.0                                | 2.6                                         |
| Guangdong-Hainan             | 838.5                                       | 210.0                                | 4.0                                         |
| Guangxi                      | 935.3                                       | 230.0                                | 4.1                                         |
| Guangzhou                    | 196.9                                       | —                                    | —                                           |
| Guiyang                      | 272.6                                       | —                                    | —                                           |
| Guizhou                      | 895.2                                       | 170.0                                | 5.3                                         |
| Hebei-Anhui-Henah            | 3147.4                                      | 470.0                                | 6.7                                         |
| Hubei                        | 610.1                                       | 180.0                                | 3.4                                         |
| Hunan                        | 422.6                                       | 160.0                                | 2.6                                         |
| North-east Mongolia-Ningxia  | 744.9                                       | 1200.0                               | 0.6                                         |
| Jiangsu                      | 5030.3                                      | 100.0                                | 50.3                                        |
| Jiangxi                      | 312.7                                       | 160.0                                | 2.0                                         |
| Heilongjiang-Jilin-Liaoning  | 2591.8                                      | 780.0                                | 3.3                                         |
| Shanghai                     | 504.7                                       | 6.2                                  | 81.6                                        |
| Shenyang                     | 170.6                                       | —                                    | —                                           |
| Shaanxi-Gansu                | 892.7                                       | 640.0                                | 1.4                                         |
| Shandong                     | 947.2                                       | 150.0                                | 6.3                                         |
| Shanxi                       | 754.4                                       | 150.0                                | 5.0                                         |
| Sichuan                      | 3031.5                                      | 570.0                                | 5.3                                         |
| Taiyuan                      | 208.0                                       | —                                    | —                                           |
| Tianjin                      | 237.7                                       | 11.3                                 | 21.0                                        |
| Tibet-Qinghai-Xinjiang Uygur | 350.3                                       | 2800.0                               | 0.1                                         |
| Wuhan                        | 160.3                                       | —                                    | —                                           |
| Yunan                        | 919.5                                       | 390.0                                | 2.4                                         |
| Zhejiang                     | 545.9                                       | 100.0                                | 5.5                                         |
| <b>India</b>                 | <b>4951.5</b>                               | <b>3288.0</b>                        | <b>1.5</b>                                  |
| Andhra Pradesh               | 213.0                                       | 275.0                                | 0.8                                         |
| West Bengal                  | 278.5                                       | 88.7                                 | 3.1                                         |
| Bihar                        | 316.3                                       | 173.8                                | 1.8                                         |
| Bombay                       | 207.1                                       | 0.2                                  | 345.2                                       |
| Calcutta                     | 62.2                                        | 0.2                                  | 311.0                                       |
| Delhi                        | 73.5                                        | 1.5                                  | 49.0                                        |
| North-east Highlands-Assam   | 455.9                                       | 254.8                                | 1.8                                         |
| Gujarat                      | 167.3                                       | 196.0                                | 0.9                                         |
| Haryana                      | 193.9                                       | 44.2                                 | 4.4                                         |
| Karnataka-Goa                | 160.7                                       | 195.5                                | 0.8                                         |
| Kerala                       | 106.3                                       | 38.8                                 | 2.7                                         |
| Madras                       | 13.1                                        | 0.2                                  | 77.0                                        |
| Maharashtra-Daman-Diu        | 320.4                                       | 307.7                                | 1.0                                         |
| Madhya Pradesh               | 378.3                                       | 443.4                                | 0.9                                         |
| Orissa                       | 219.0                                       | 155.7                                | 1.4                                         |
| Punjab-Chandigarh            | 160.2                                       | 50.4                                 | 3.2                                         |
| Rajasthan                    | 130.0                                       | 342.2                                | 0.4                                         |
| Tamil Nadu-Pondichery        | 227.1                                       | 130.0                                | 1.7                                         |
| Uttar Pradesh                | 460.5                                       | 294.4                                | 1.6                                         |
| Jammu-Kashmir-Himachal       | 808.2                                       | —                                    | —                                           |

\* Sources: Kaplan *et al.* (1982) and GOI (1993).

TABLE 9. Sectoral emission of SO<sub>2</sub> in Asian regions/countries in 1990

| Region/country         | Sectoral shares (%) |             |             |             | Total emission (tons) |
|------------------------|---------------------|-------------|-------------|-------------|-----------------------|
|                        | Industrial          | Domestic    | Power       | Transport   |                       |
| <b>North-east Asia</b> | <b>52.4</b>         | <b>18.3</b> | <b>26.0</b> | <b>3.3</b>  | <b>29 694.0</b>       |
| China                  | 53.7                | 18.7        | 26.2        | 1.4         | 26 373.9              |
| Hong Kong              | 4.6                 | 3.3         | 87.0        | 19.2        | 138.5                 |
| Japan                  | 30.8                | 9.6         | 23.7        | 35.8        | 852.1                 |
| South Korea            | 45.5                | 21.2        | 25.3        | 8.0         | 1770.2                |
| Mongolia               | 1.3                 | 3.2         | 94.9        | 0.6         | 46.7                  |
| Taiwan                 | 69.7                | 8.6         | 9.1         | 12.6        | 512.6                 |
| <b>South Asia</b>      | <b>38.1</b>         | <b>11.5</b> | <b>42.1</b> | <b>8.3</b>  | <b>5742.2</b>         |
| Bangladesh             | 32.3                | 43.0        | 10.1        | 14.6        | 87.4                  |
| Bhutan                 | 9.5                 | 85.7        | —           | 4.8         | 2.1                   |
| India                  | 36.4                | 10.2        | 45.8        | 7.5         | 4951.5                |
| Myanmar                | 3.6                 | 93.7        | 1.4         | 1.4         | 22.1                  |
| Nepal                  | 9.6                 | 82.2        | 3.9         | 4.3         | 28.0                  |
| Pakistan               | 55.3                | 9.6         | 22.1        | 13.0        | 620.6                 |
| Sri Lanka              | 37.3                | 31.4        | 0.7         | 30.7        | 30.5                  |
| <b>South-east Asia</b> | <b>29.6</b>         | <b>11.6</b> | <b>46.8</b> | <b>12.0</b> | <b>2710.0</b>         |
| Brunei                 | 5.1                 | 7.7         | 51.3        | 35.9        | 3.9                   |
| Cambodia               | 45.4                | 21.9        | 26.5        | 6.1         | 19.6                  |
| Indonesia              | 33.5                | 22.9        | 26.9        | 16.7        | 691.1                 |
| Laos                   | 41.4                | 51.7        | 3.4         | 3.4         | 4.2                   |
| Malaysia               | 27.6                | 7.1         | 59.7        | 5.4         | 214.3                 |
| Philippines            | 36.8                | 4.7         | 53.1        | 5.5         | 414.9                 |
| Singapore              | 34.3                | 0.2         | 59.7        | 5.4         | 190.9                 |
| Thailand               | 22.9                | 7.0         | 55.6        | 14.5        | 1036.9                |
| Vietnam                | 33.9                | 36.5        | 18.1        | 11.5        | 134.4                 |
| <b>Asia (total)</b>    | <b>48.6</b>         | <b>16.8</b> | <b>29.9</b> | <b>4.7</b>  | <b>38 146.2</b>       |

wide variations (from 0.17–190 tons/km<sup>2</sup>) across countries/areas in the region. Eight countries (China, Hong Kong, Japan, South Korea, Philippines, Singapore, Taiwan and Thailand) were found to have above average emission densities. These countries had higher emission densities than those of the U.S.A., the United Kingdom, Japan, Germany and France. Given the high economic growth rates of most countries in the region, it can be expected that the emission density will become significantly higher in the coming decades if the present structure of production continues. The emission density figures at the national level could, however, be misleading in that they could mask the large variations in densities that exist across areas within a particular country. This is illustrated in Table 8, in the cases of China and India. Although the national level emission densities of China and India appear relatively low (2.8 and 1.5 tons/km<sup>2</sup>, respectively), there were at least 11 sub-country regions in China and four in India for which the emission density was above 3 tons/km<sup>2</sup>. These figures were higher than those of all other countries and sub-country regions of Asia except Singapore, Hong Kong, Taiwan and South Korea.

#### 4.2. SECTORAL EMISSIONS

Table 9 presents the sectoral emission of SO<sub>2</sub> of Asian countries in 1990. The industrial sector accounted for the greatest share (55%) of the total emission of SO<sub>2</sub> in the whole

TABLE 10. Emissions of SO<sub>2</sub> by fuel type in Asian regions/countries in 1990

| Region/country         | Share by fuel type (%) |             |            |               | Total emission (tons) |
|------------------------|------------------------|-------------|------------|---------------|-----------------------|
|                        | Coal                   | Oil         | Gas        | Other solids* |                       |
| <b>North-east Asia</b> | <b>89.6</b>            | <b>10.3</b> | —          | —             | <b>29 694.0</b>       |
| China                  | 96.9                   | 3.1         | —          | —             | 26 373.9              |
| Hong Kong              | 75.1                   | 24.8        | —          | 0.1           | 138.5                 |
| Japan                  | 23.7                   | 76.3        | —          | 0.1           | 852.1                 |
| South Korea            | 32.2                   | 67.7        | —          | 0.2           | 1770.2                |
| Mongolia               | 90.8                   | 7.1         | —          | 2.1           | 46.7                  |
| Taiwan                 | 37.8                   | 62.1        | 0.1        | —             | 512.6                 |
| <b>South Asia</b>      | <b>81.4</b>            | <b>16.7</b> | <b>0.1</b> | <b>1.8</b>    | <b>2710.0</b>         |
| Bangladesh             | 14.2                   | 34.6        | 1.3        | 49.9          | 87.4                  |
| Bhutan                 | 4.8                    | 9.5         | —          | 85.7          | 2.1                   |
| India                  | 69.2                   | 23.4        | 0.5        | 6.9           | 4951.5                |
| Myanmar                | 1.8                    | 5.4         | 0.9        | 91.9          | 22.1                  |
| Nepal                  | 3.6                    | 16.4        | —          | 80.0          | 28.0                  |
| Pakistan               | 41.2                   | 54.5        | 0.3        | 4.1           | 620.6                 |
| Sri Lanka              | 0.3                    | 69.5        | —          | 30.2          | 30.5                  |
| <b>South-east Asia</b> | <b>81.4</b>            | <b>16.7</b> | <b>0.1</b> | <b>1.8</b>    | <b>2710.0</b>         |
| Brunei                 | —                      | 44.7        | 55.3       | —             | 3.9                   |
| Cambodia               | 52.0                   | 26.0        | —          | 21.9          | 19.6                  |
| Indonesia              | 6.6                    | 74.9        | —          | 18.5          | 691.1                 |
| Laos                   | —                      | 31.0        | —          | 69.0          | 4.2                   |
| Malaysia               | 7.1                    | 90.0        | 0.4        | 2.5           | 214.3                 |
| Philippines            | 24.3                   | 75.7        | —          | —             | 414.9                 |
| Singapore              | 0.1                    | 99.9        | 0.1        | —             | 190.9                 |
| Thailand               | 49.0                   | 46.8        | —          | 4.2           | 1036.9                |
| Vietnam                | 20.8                   | 47.6        | —          | 31.6          | 134.4                 |
| <b>Asia</b>            | <b>81.4</b>            | <b>16.7</b> | <b>0.1</b> | <b>1.8</b>    | <b>38 146.2</b>       |

\* The other solids include fuel wood and biomass fuels (such as plant residue and animal waste).

of Asia, followed by the power generation (22%), domestic (18%) and transport (5%) sectors, respectively. At the individual country level, the industrial sector had the largest share of total national SO<sub>2</sub> emissions in China, Japan, South Korea, Taiwan, Indonesia, Pakistan, Sri Lanka, Cambodia and Laos. Power generation had the largest share of national emissions in India, Hong Kong, Malaysia, Philippines, Thailand, Singapore, Brunei and Mongolia. Together, the industrial and power generation sectors contributed about 77% of the total SO<sub>2</sub> emission in Asia.

#### 4.3. EMISSIONS BY FUEL TYPE

Emissions of SO<sub>2</sub> from fuel are shown in Table 10. Coal (of different varieties, i.e. brown coal, hard coal and derived coke) was the largest source of emission in 1990, accounting for about 81% of the total emission from the region, followed by oil (17%). At the individual country level, coal use was the biggest source of emissions in China, India, Hong Kong, Thailand, Cambodia and Mongolia, while oil was the main source in Japan, South Korea, Taiwan, Pakistan, Sri Lanka, Indonesia, Malaysia, Philippines, Singapore and Vietnam.

## 5. Conclusion and final remarks

From the present analysis, the total emission of SO<sub>2</sub> in Asia was estimated to be about 38 million tons in 1990. About 78% of these emissions originated from countries in north-east Asia (China, Hong Kong, Japan, Mongolia, South Korea and Taiwan). Coal use accounted for over 80% of the total emissions in Asia as a whole and about 90% of the total emissions in north-east Asia. Due to rapid economic growth and the extensive use of coal in the energy sector, SO<sub>2</sub> emissions are expected to be much higher in the coming decades if effective mitigation measures are not adopted.

In terms of sectoral contributions to SO<sub>2</sub> emissions, the industrial sector was a significant source of total emissions from Asia as a whole and that from north-east Asia, while it was the power sector in south and south-east Asian regions. Clearly, the findings show that, in terms of mitigation strategies, the industrial sector deserves more attention in north-east Asia while options to control the power sector may be more important in south and south-east Asia.

There seems to exist a large potential for the mitigation of SO<sub>2</sub> emissions through efficiency improvements in the industrial and power sectors of high emission countries like China and India. However, these remain mostly untapped because substantial investments are needed. Additional financial investment and associated pricing implications are the problems confronted by the policy makers in most countries in the region. We intend to examine the mitigation potential of various energy efficiency and emission control options as well as the associated costs in a subsequent paper.

This study reports the findings of an analysis based on an energy and socio-economic database established in the framework of the regional project entitled "Acid Rain and Emissions Reduction in Asia" funded by the Asian Development Bank (ADB). However, the views expressed in the paper are those of the authors only and not necessarily of ADB. Collin Green and Wesley K. Foell of Resource Management Associates of Madison (RMA), Madison, U.S.A. and Markus Amann of International Institute for Applied Systems (IIASA), Laxenburg, Austria provided useful inputs to set up the database.

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